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B.Tech. I Semester Examination, 2025

Regulation: R20 | University: JNTUK

Engineering Thermodynamics (21ME32)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Basic Concepts & Zeroth Law]

- a) Distinguish between closed, open and isolated systems with examples.
- b) Explain the zeroth law of thermodynamics and its applications.

UNIT - 2 [First Law of Thermodynamics]

- a) Derive the steady flow energy equation and apply it to a nozzle.
- b) Derive the steady flow energy equation and apply it to a nozzle.

UNIT - 3 [Second Law, Entropy & Availability]

- a) Explain the equivalence of Kelvin Planck and Clausius statements.
- b) Prove that the entropy of the universe always increases.

UNIT - 4 [Properties of Pure Substances & Gas Mixtures]

- a) Explain the Mollier chart and its engineering uses.
- b) Explain the Mollier chart and its engineering uses.

UNIT - 5 [Vapour & Gas Power Cycles]

- a) Compare Otto and Diesel cycles with p-v and T-s diagrams.
- b) Explain the Rankine cycle with a T-s diagram.

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